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CAMBRIDGE ADDITIONAL MATHEMATICS (0606) • UNIT 4.01

Logarithmic & Exponential Functions

PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Logarithmic & Exponential Functions Practice Exercises

1 MCQ

Solve the exponential equation $2^{x+2} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{2\ln 2}{\ln 5 - \ln 2}$

B $x = \frac{2\ln 5}{\ln 5 - \ln 2}$

C $x = \frac{\ln 2}{2(\ln 5 - \ln 2)}$

D $x = \frac{2}{\ln 5 - \ln 2}$

2 MCQ

Solve the exponential equation $2^{x+3} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{3\ln 2}{\ln 5 - \ln 2}$

B $x = \frac{3\ln 5}{\ln 5 - \ln 2}$

C $x = \frac{3}{\ln 5 - \ln 2}$

D $x = \frac{\ln 2}{3(\ln 5 - \ln 2)}$

3 MCQ

Solve the exponential equation $2^{x+4} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{4\ln 2}{\ln 5 - \ln 2}$

B $x = \frac{\ln 2}{4(\ln 5 - \ln 2)}$

C $x = \frac{4\ln 5}{\ln 5 - \ln 2}$

D $x = \frac{4}{\ln 5 - \ln 2}$

4 MCQ

Solve the exponential equation $2^{x+5} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{5\ln 2}{\ln 5 - \ln 2}$

B $x = \frac{5\ln 5}{\ln 5 - \ln 2}$

C $x = \frac{5}{\ln 5 - \ln 2}$

D $x = \frac{\ln 2}{5(\ln 5 - \ln 2)}$

5 MCQ

Solve the exponential equation $2^{x+6} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{6\ln 5}{\ln 5 - \ln 2}$

B $x = \frac{6\ln 2}{\ln 5 - \ln 2}$

C $x = \frac{6}{\ln 5 - \ln 2}$

D $x = \frac{\ln 2}{6(\ln 5 - \ln 2)}$

6 MCQ

Solve the exponential equation $2^{x+7} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{7 \ln 2}{\ln 5 - \ln 2}$

B $x = \frac{\ln 2}{7(\ln 5 - \ln 2)}$

C $x = \frac{7 \ln 5}{\ln 5 - \ln 2}$

D $x = \frac{7}{\ln 5 - \ln 2}$

7 MCQ

Solve the exponential equation $2^{x+8} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{8 \ln 2}{\ln 5 - \ln 2}$

B $x = \frac{8 \ln 5}{\ln 5 - \ln 2}$

C $x = \frac{\ln 2}{8(\ln 5 - \ln 2)}$

D $x = \frac{8}{\ln 5 - \ln 2}$

8 MCQ

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 3$. What is the value of x?

A 12

B 3

C 6

D 14

9 MCQ

Find the root(s) satisfying the exponential equation $2^{x+3} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{3 \ln 2}{\ln 5 - \ln 2}$

B $x = \frac{3 \ln 5}{\ln 5 - \ln 2}$

C $x = \frac{3}{\ln 5 - \ln 2}$

D $x = \frac{\ln 2}{3(\ln 5 - \ln 2)}$

10 MCQ

Calculate the solution set for the exponential equation $2^{x+4} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{4 \ln 2}{\ln 5 - \ln 2}$

B $x = \frac{\ln 2}{4(\ln 5 - \ln 2)}$

C $x = \frac{4 \ln 5}{\ln 5 - \ln 2}$

D $x = \frac{4}{\ln 5 - \ln 2}$

11 MCQ

Obtain the value(s) satisfying the exponential equation $2^{x+5} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{5 \ln 5}{\ln 5 - \ln 2}$

B $x = \frac{5}{\ln 5 - \ln 2}$

C $x = \frac{\ln 2}{5(\ln 5 - \ln 2)}$

D $x = \frac{5 \ln 2}{\ln 5 - \ln 2}$

12 MCQ

Work out the solution to the exponential equation $2^{x+6} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{6}{\ln 5 - \ln 2}$

B $x = \frac{\ln 2}{6(\ln 5 - \ln 2)}$

C $x = \frac{6 \ln 5}{\ln 5 - \ln 2}$

D $x = \frac{6 \ln 2}{\ln 5 - \ln 2}$

13 MCQ

Determine the solution to the exponential equation $2^{x+7} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{7 \ln 5}{\ln 5 - \ln 2}$

B $x = \frac{7 \ln 2}{\ln 5 - \ln 2}$

C $x = \frac{7}{\ln 5 - \ln 2}$

D $x = \frac{\ln 2}{7(\ln 5 - \ln 2)}$

14 MCQ

Find the root(s) satisfying the exponential equation $2^{x+8} = 5^x$, giving x in terms of common logarithms.

A $x = \frac{8 \ln 2}{\ln 5 - \ln 2}$

B $x = \frac{8 \ln 5}{\ln 5 - \ln 2}$

C $x = \frac{\ln 2}{8(\ln 5 - \ln 2)}$

D $x = \frac{8}{\ln 5 - \ln 2}$

15 MCQ

Find the root(s) satisfying the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 4$. What is the value of x?

A 8

B 16

C 4

D 18

16 MCQ

Calculate the solution set for the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 5$. What is the value of x?

A 5

B 22

C 10

D 20

17 MCQ

Obtain the value(s) satisfying the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 6$. What is the value of x?

A 12

B 24

C 6

D 26

18 MCQ

Work out the solution to the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 7$. What is the value of x?

A 14

B 30

C 28

D 7

19 MCQ

Determine the solution to the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 8$. What is the value of x?

A 16

B 8

C 32

D 34

20 MCQ

Find the root(s) satisfying the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 9$. What is the value of x?

A 9

B 38

C 18

D 36

21 MCQ

Variables x and y satisfy $y = Ae^{bx}$. When $\ln y$ is plotted against x (Experiment #1), a straight line with gradient 2 and vertical intercept 1 is obtained. Find the exact value of y when $x = 2$.

A e^5

B $1e^2$

C e^4

D 5

22 MCQ

Variables x and y satisfy $y = Ae^{bx}$. When $\ln y$ is plotted against x (Experiment #2), a straight line with gradient -3 and vertical intercept 1 is obtained. Find the exact value of y when $x = 3$.

A e^0

B e^{-8}

C -8

D $1e^{-3}$

23 MCQ

Variables x and y satisfy $y = Ae^{bx}$. When $\ln y$ is plotted against x (Experiment #3), a straight line with gradient 4 and vertical intercept 1 is obtained. Find the exact value of y when $x = 4$.

A 17

B e^{17}

C $1e^4$

D e^8

24 MCQ

Variables x and y satisfy $y = Ae^{bx}$. When $\ln y$ is plotted against x (Experiment #4), a straight line with gradient -5 and vertical intercept 1 is obtained. Find the exact value of y when $x = 5$.

A -24

B $1e^{-5}$

C e^{-24}

D e^{0}

25 MCQ

Variables x and y satisfy $y = Ae^{bx}$. When $\ln y$ is plotted against x (Experiment #5), a straight line with gradient 6 and vertical intercept 1 is obtained. Find the exact value of y when $x = 2$.

A e^{8}

B e^{13}

C $1e^{6}$

D 13